

Sequence pairs

A sequence pair is two permutations of the modules, often called the positive and negative sequences, Γ -plus and Γ -minus

The idea

- If module A appears before B in both sequences, A is left of B
- If A appears before B in Γ -plus but after B in Γ -minus
- Horizontal and vertical constraint graphs yield x and y via longest paths
- Soft sizing and rotation can be layered on later; first get hard rectangles right

Browser lab track

- Open **sequence-pair**
- Load starter Γ -plus and Γ -minus
- Highlight one pair of modules and read the implied relation, left-of or above
- Pack and inspect deadspace under the fixed outline
- Swap two symbols in one sequence and watch relations flip
- Then implement packing from sequences in Track A

Implement track

- Represent Γ -plus and Γ -minus as lists of module ids
- Build constraint edges, compute coordinates
- Report deadspace
- Keep permutations as permutations, no duplicate ids
- Deterministic packing helps goldens

Pitfalls

- Mixing up which sequence pair condition means “left of” versus “above” yields mirrored
- Forgetting transitive longest-path compaction leaves unnecessary whitespace
- And sequence pairs can encode non-slicing packings

Your turn

- Pack A–E from a sequence pair and verify legality
- Next you'll search the space of representations with simulated annealing